

SIH26034 - Legal Metrology Automated Compliance System

Definitive MVP Scope, MoSCoW Prioritization & Explicit Cut List for 13 September 2026 Deadline

1. Strategic Context & Hackathon Realities

- **Target Competition:** Smart India Hackathon 2026 (Problem Statement ID: SIH26034)
- **Ministry / Department:** Ministry of Consumer Affairs, Food & Public Distribution / Department of Consumer Affairs (DoCA)
- **Execution Window:** 07 September 2026 to 13 September 2026 (6 Days Remaining)
- **Team Size:** 6-Member Multidisciplinary Engineering Team
- **Guiding Engineering Axiom:**

"A complete, bug-free, legally unassailable end-to-end vertical slice always triumphs over a wide, half-broken horizontal architecture with mock data."

The scope defined herein is **FROZEN**. No features outside P0 may be added until the P0 pipeline is passing 100% of integration test suites and has executed a complete end-to-end live run.

2. MoSCoW Prioritization Matrix

+-----+ -----+ PRIORITY LEVEL DESCRIPTION & EXECUTION STATUS +-----+ -----+	
P0: MUST-HAVE	Mandatory core pipeline. Failure to deliver any item = Project Incompletion.
(MVP Freeze)	Deadline for code complete: 10 September 2026 (Day 4).
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P1: SHOULD-HAVE	High-value differentiators. Implemented only after P0 integration passes.
(Stretch Goals)	Target delivery: 11 September 2026 (Day 5).
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P2: COULD-HAVE	Post-hackathon architectural roadmap for state-wide scaling.
(Roadmap)	Documented in architecture blueprints; omitted from demo code.
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P3: WON'T-HAVE	Explicitly blacklisted features, hazardous designs, and scope-creep traps.
(Cut List)	Zero development time allocated. Rejected by architectural consensus.
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3. Detailed Scope Breakdown

3.1 P0: Must-Have (MVP Frozen Scope)

1. Physical Label Ingestion & Optical Quality Gate:

- Ingestion of high-resolution packaging images ($\geq 1920 \times 1080$).
- Automated quality gating: Laplacian blur variance ($\sigma^2 \geq 150$), specular glare threshold ($\leq 3\%$), and perspective skew limit ($\leq 15^\circ$).

2. Deterministic Fiducial Calibration Engine:

- Detection of ArUco 4x4 50 mm reference marker (with fallback to ₹5 standard coin).
- Planar homography perspective rectification (`cv2.warpPerspective`) to eliminate camera tilt distortion.
- Derivation of exact pixel-to-millimeter scaling ratio (`px_to_mm`).

3. Apache-2.0 Permissive OCR & Text Detection Pipeline:

- DBNet++ / PaddleDetection for multi-oriented text polygon detection.
- PaddleOCR PP-OCRv4 for high-accuracy text recognition on English statutory declarations.
- Connected-components bounding analysis for character x-height measurement.

4. Core Legal Metrology Rule Engine (Packaged Commodities Rules, 2011):

- **Rule 6(1)(a):** Name and address of manufacturer, packer, or importer.
- **Rule 6(1)(h) & Table-I:** Net quantity declaration and minimum font height validation based on calculated Principal Display Panel (PDP) area:
 - $\text{Area} \leq 50 \text{ cm}^2$: $\geq 1.0 \text{ mm}$
 - $50 < \text{Area} \leq 100 \text{ cm}^2$: $\geq 1.5 \text{ mm}$
 - $100 < \text{Area} \leq 500 \text{ cm}^2$: $\geq 2.0 \text{ mm}$ (Weight $\leq 200\text{g}$) / $\geq 4.0 \text{ mm}$ (Weight $> 200\text{g}$)
 - $500 < \text{Area} \leq 2500 \text{ cm}^2$: $\geq 4.0 \text{ mm}$
 - $\text{Area} > 2500 \text{ cm}^2$: $\geq 6.0 \text{ mm}$ (Official Gazette G.S.R. 629(E) boundary).
- **Rule 6(1)(e):** Maximum Retail Price (MRP) declaration including mandatory "(inclusive of all taxes)" clause.
- **Rule 6(1)(k):** Unit Sale Price (USP) validation:
 - Packaged commodities $< 1 \text{ kg}/1 \text{ L}$: Declared per gram / per milliliter.
 - Packaged commodities $\geq 1 \text{ kg}/1 \text{ L}$: Declared per kilogram / per liter.
 - Mathematical accuracy cross-check: $|\text{USP}_{\text{calc}} - \text{USP}_{\text{declared}}| \leq 0.01$
- **Rule 6(1)(n):** Consumer care name, address, telephone number, and email.
- **Rule 6(1)(p):** Country of origin declaration (especially for imported goods).

5. Interactive Officer Adjudication Canvas:

- Split-view web application: Calibrated zoom/pan image canvas with visual bounding box overlays on the left; structured statutory rule ledger on the right.
- Pixel loupe tool with real-time millimeter readout.
- Human-in-the-Loop approval / override workflow with mandatory justification logging.

6. Statutory Legal Notice & Section 63 BSA 2023 Generator:

- Automated compilation of Form-1 / Form-2 Legal Notice under Section 36(1) of the Legal Metrology Act, 2009.
- Generation of Section 63 Bharatiya Sakshya Adhiniyam, 2023 (BSA 2023) Electronic Evidence Certificate embedding raw image SHA-256 hash, device metadata, and officer signature.

- Archival PDF/A export with embedded evidence exhibits.

7. E-Commerce Single-Listing Compliance Inspector:

- Ingestion of product listing URL, HTML DOM snapshot, and primary gallery product packaging images.
- Rule 6(10) declaration completeness check (Name of packer, country of origin, net qty, MRP).
- Rule 6(10A) Country of Origin search/filter compliance check (G.S.R. 128(E) 2026 mandate).
- Discrepancy detector: Cross-referencing webpage textual claims against OCR text read from packaging image.

8. Tamper-Evident SHA-256 Audit Trail:

- Append-only cryptographic ledger chaining all inspection steps and officer actions.

3.2 P1: Should-Have (Stretch Scope - Only if P0 Completed by Day 4)

1. **Batch CSV Ingestion Queue:** Uploading a CSV of 100 e-commerce URLs for automated asynchronous processing.
2. **1D/2D Barcode & QR Parsing:** Pyzbar / OpenCV decoding of EAN-13 barcodes to verify prefix validity against GS1 India prefix registries (890).
3. **Multi-Lingual OCR Baseline:** Hindi Devanagari text recognition for dual-language labels (English + Hindi).

3.3 P2: Could-Have (Future Product Roadmap)

1. Native Android APK with live WebRTC/AR camera guidance overlays.
2. Enterprise SSO integration with Government Parichay / Jan Parichay identity services.
3. Direct automated API integration with National Consumer Helpline (NCH) grievance portal.

3.4 P3: Won't-Have (The Explicit Cut List & Non-Goals)

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CUT ITEM	RATIONALE FOR EXCLUSION
+-----+	

-----+ 1. Mass E-Commerce Web Scraping Mass scraping triggers IP bans, CAPTCHAs, and violates TOS. Scope is strictly restricted to single URL / DOM / gallery image ingestion. +-----+ -----+	
2. Autonomous AI Notice Dispatch Unconstitutional and illegal under Indian jurisprudence. Fines and notices require human officer adjudication sign-off. +-----+ -----+	
3. Monocular Depth Estimation Single uncalibrated images cannot resolve physical mm without Without Reference Fiducials scale ambiguity. Legally inadmissible in court. +-----+ -----+	
4. AGPL-3.0 Models (YOLOv8/v11) Viral copyleft license endangers government intellectual property. Permissive Apache-2.0 alternatives chosen. +-----+ -----+	
5. Physical Scale / Weight Testing Software cannot weigh physical goods. Net weight compliance Verification is checked strictly against printed on-pack declarations. +-----+ -----+	
6. Chemical / FSSAI Ingredient Outside Legal Metrology Department statutory jurisdiction. Adulteration Analysis Scope strictly bounded to Legal Metrology (PC) Rules, 2011. +-----+ -----+	
7. Proprietary Cloud Vision APIs Paid commercial APIs (Google Cloud Vision, AWS Rekognition) create recurring government cost dependencies and vendor lock. +-----+ -----+	